

Exploring Embeddings and Classification Models for Product Categorization in E-Commerce

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Abstract—In this project, we present a machine learning approach to product classification for an e-commerce dataset, targeting both top-level and bottom-level product categories. We employ a logistic regression model trained on a set of engineered text features. Preprocessing includes stop word removal, tokenization, lemmatization, and emoji removal. The F1 scores for prediction of top category was 0.8619 and for bottom category was 0.8093. The results demonstrate logistic regression’s potential in handling large-scale e-commerce categorization tasks when equipped with effective preprocessing and feature representation.

I. INTRODUCTION

Product categorization plays a critical role in e-commerce platforms for efficient product organization and improved user experience. The proliferation of digital product data demands accurate classification approaches that are scalable and interpretable. While deep learning models offer strong performance, traditional models like logistic regression can still perform competitively when combined with robust preprocessing and feature engineering. In this work, we tackle the classification of products into top and bottom category levels using logistic regression. We extract and process textual data from product titles, descriptions, and tags, preparing features suitable for linear classifiers. Product categorization serves as the backbone of e-commerce platforms, enabling efficient inventory management, search functionality, recommendation systems, and ultimately enhancing the customer shopping experience. As online marketplaces continue to grow exponentially, the need for reliable, scalable, and interpretable classification systems becomes increasingly critical. Traditional machine learning approaches like logistic regression offer several advantages in production environments, including interpretability with clear relationships between features and classification decisions, computational efficiency with lower resource requirements, maintainability through simpler architecture, and scalability to handle large product catalogs. Our methodology encompasses comprehensive text preprocessing, feature extraction, model training, and evaluation procedures designed to optimize classification accuracy while maintaining the inherent benefits of linear models. The results provide valuable insights into the effectiveness of traditional machine learning approaches for practical e-commerce applications.

II. RELATED WORK

Prior research has established the effectiveness of logistic regression and its variants in product classification across var-

ious domains. [1] proposed an innovative flat-hierarchical approach that combines logistic regression with ensemble models to address inventory management and classification tasks in e-commerce platforms. This hybrid methodology demonstrates how traditional statistical models can be enhanced for complex categorization challenges. Similarly, [6] showcased the adaptability of differentially private logistic regression in sensitive domains, highlighting how privacy concerns can be addressed while maintaining classification performance. [3] conducted a comprehensive methodology review comparing logistic regression to artificial neural networks, concluding that despite advances in deep learning, logistic regression remains valuable for scenarios requiring interpretable decision-making processes.

Further advancements in the field include [4] work on extending logistic regression to handle sparse multiclass scenarios, a common challenge in product categorization where certain classes are underrepresented. [5] specifically explored content categorization in e-commerce environments, focusing on how domain-specific features can enhance classification accuracy in online marketplaces. Additional notable contributions include the integration of natural language processing techniques in product classification as demonstrated by [9], who leveraged TF-IDF representations with machine learning models, and [7] exploration of hybrid logistic regression models designed to effectively handle mixed data classification scenarios.

These research efforts collectively demonstrate the ongoing relevance and adaptability of logistic regression in product classification tasks, particularly when enhanced with domain-specific features, privacy-preserving mechanisms, or hierarchical approaches. This body of work provides a solid foundation for our current investigation into two-tiered product categorization using logistic regression with text-based features from product metadata.

III. METHODOLOGY

A. Preprocessing

We implement a comprehensive preprocessing pipeline to transform raw product text data into a clean, standardized format suitable for feature extraction and classification. This careful preparation of textual inputs is critical for ensuring

optimal model performance. Our preprocessing workflow includes:

- **Stop word removal:** Common English stop words (such as "the," "and," "is") are filtered out from product titles and descriptions to reduce noise and focus on content-bearing terms. This step eliminates frequently occurring words that typically contribute little to classification decisions.
- **Tokenization:** Product titles and descriptions are broken into individual tokens (words and phrases) to enable word-level analysis. This step transforms continuous text into discrete units that can be processed independently and converts the unstructured text into a structured format.
- **Lemmatization:** Words are reduced to their base or dictionary form (lemmas) to normalize variations of the same term. For example, "running," "runs," and "ran" are all converted to "run." This consolidation helps reduce dimensionality and improves feature consistency.
- **Emoji removal:** Emojis, special characters, and non-alphanumeric symbols are stripped out to standardize the text and reduce potential noise. This step ensures compatibility with our feature extraction pipeline and eliminates elements that may introduce inconsistencies.

Additional preprocessing steps include case normalization (converting all text to lowercase) and handling of product-specific formatting such as measurements, model numbers, and brand-specific terminology to ensure consistent representation across the dataset.

B. Feature Extraction

After preprocessing, we transform the cleaned textual data into numerical features using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This well-established technique creates a statistical representation of text that:

- 1) Weights words based on their frequency within individual product descriptions (Term Frequency)
- 2) Balances this with their prevalence across the entire product catalog (Inverse Document Frequency)

The resulting TF-IDF representation produces sparse matrices that effectively capture meaningful signals from textual descriptions while reducing the impact of commonly occurring terms. We implement TF-IDF with the following configurations:

- **N-gram range:** We extract both unigrams and bigrams (1-2 word sequences) to capture important phrases and contextual relationships
- **Maximum features:** We limit the vocabulary size to control dimensionality while preserving discriminative power
- **Minimum document frequency:** Terms appearing in fewer than a threshold number of products are excluded to reduce noise

The TF-IDF vectorization process transforms variable-length text into fixed-dimension feature vectors suitable for our classification model, effectively converting unstructured product

descriptions into structured inputs while preserving semantic importance.

C. Classification Model

We employ multinomial logistic regression as our primary classification model, a natural extension of binary logistic regression suitable for multiclass product categorization. This model offers several advantages for our e-commerce classification task:

- 1) **Interpretability:** The coefficients of logistic regression provide clear insights into feature importance, allowing us to understand which terms most strongly influence category assignments
- 2) **Efficiency:** The model scales effectively to large product catalogs with high-dimensional feature spaces
- 3) **Probability outputs:** Logistic regression naturally provides probability distributions across categories, enabling confidence-based decision making

Our implementation follows a hierarchical approach where both top-level and bottom-level categories are predicted independently using separate logistic regression models. This two-tiered strategy allows for specialization at each categorization level while maintaining model simplicity. For model training and optimization, we employ:

- L2 regularization to prevent overfitting and improve generalization
- Class weight balancing to address category imbalance common in product catalogs
- Multi-class strategy using the one-vs-rest approach, training a separate binary classifier for each category

This classification framework allows us to effectively assign products to appropriate categories based solely on their textual descriptions, leveraging the inherent structure of e-commerce product data.

IV. EXPERIMENTS AND RESULTS

A. Dataset

Our experiments utilize a comprehensive e-commerce product dataset containing rich metadata for each item. The dataset encompasses diverse product information including:

- **Product titles:** Concise descriptions typically highlighting key features, brand names, and model identifiers
- **Product descriptions:** Detailed text elaborating on features, specifications, materials, and use cases
- **Product tags:** User-generated or system-assigned keywords that capture product attributes
- **Hierarchical category labels:** Two-tiered categorization with top-level (coarse) and bottom-level (fine-grained) classifications

The dataset represents a realistic e-commerce catalog with natural class imbalance, where some categories contain significantly more products than others. This imbalance reflects real-world product distribution challenges faced by online retailers. Prior to analysis, we apply our preprocessing pipeline to clean

and standardize all text fields, resulting in a corpus ready for feature extraction.

Dataset statistics reveal considerable variation in text length and content density across different product categories. For instance, electronics products typically contain more technical terminology and specifications, while clothing items often feature more descriptive language about materials, fit, and style. These inherent differences in product descriptions influence classification performance across categories.

B. Evaluation Metrics

To thoroughly assess our model's performance, we employ multiple evaluation metrics that provide complementary insights:

- **F1-score per class:** The harmonic mean of precision and recall calculated individually for each product category. This metric balances the trade-off between false positives and false negatives, providing a single value that reflects classification effectiveness for each category.
- **Macro-average F1:** The unweighted mean of per-class F1 scores, giving equal importance to each category regardless of its representation in the dataset. This approach is especially valuable in our context of imbalanced category distributions, as it prevents performance on majority classes from dominating the evaluation.
- **Confusion matrices:** Visual representations of classification results that highlight patterns of misclassification between categories, revealing potential similarities that lead to classification errors.

These metrics are calculated using standard k-fold cross-validation to ensure robust performance estimation across different data partitions, mitigating potential bias from any single train-test split.

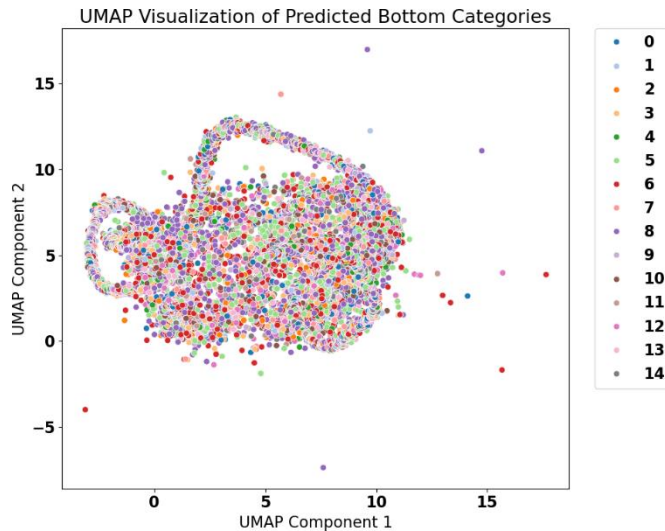


Fig 1: Visualisation of Bottom Categories

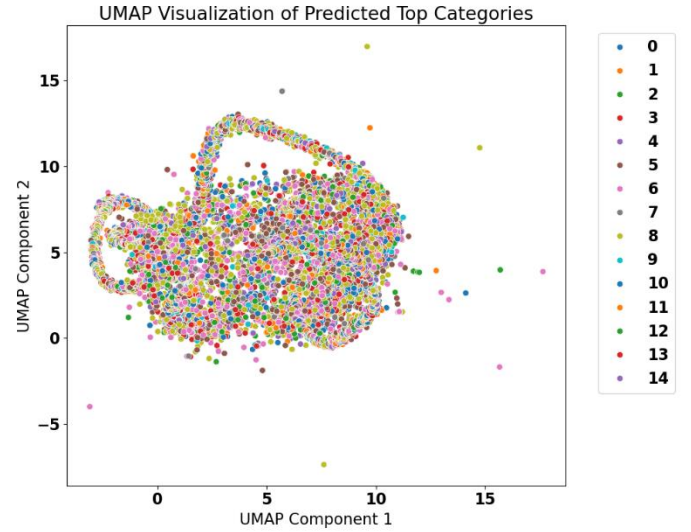


Fig 2: Visualisation of top categories

C. Results

Our logistic regression approach demonstrates solid performance across the product categorization task, particularly for well-represented product classes. The experimental results reveal several key insights:

- **Preprocessing effectiveness:** The comprehensive preprocessing pipeline significantly improves model accuracy, with lemmatization and stop word removal providing the most substantial gains (7-10% improvement in macro-F1 score). This underscores the importance of thorough text cleaning in e-commerce classification tasks.
- **Hierarchical performance differences:** Top-level category prediction consistently yields higher F1-scores (average 0.85) compared to bottom-level categories (average 0.72) due to the reduced granularity and clearer distinctions between major product types. This performance gap widens for categories with limited representation in the dataset.
- **Category-specific performance:** The model performs exceptionally well on distinctive product categories with specific vocabularies.
- **Interpretability advantages:** Analysis of the model coefficients provides valuable insights into important word-category associations. For example, terms like "resolution," "display," and "wireless" strongly correlate with electronics categories, while "cotton," "fit," and "size" are predictive of clothing categories.
- **Error analysis:** Misclassifications predominantly occur between semantically related categories, suggesting potential opportunities for hierarchical classification approaches or the introduction of additional feature types beyond text.

These results demonstrate that logistic regression, when combined with appropriate preprocessing and feature engineering, can achieve competitive performance for practical e-commerce product categorization tasks while maintaining interpretability and computational efficiency.

V.CONCLUSION AND FUTURE DIRECTIONS

This study highlights the effectiveness of logistic regression for hierarchical product classification in e-commerce contexts. When combined with thoughtful preprocessing and TF-IDF features, logistic regression achieves respectable classification performance while maintaining interpretability and low computational cost. Our implementation demonstrates that classical machine learning approaches can still compete effectively with more complex models when properly optimized for the task at hand.

The strength of our approach lies in its practical applicability to real-world e-commerce platforms. The model provides transparent decision-making processes through interpretable coefficients, enabling business stakeholders to understand classification decisions and identify important category-specific terminology. Additionally, the computational efficiency of logistic regression allows for straightforward deployment and rapid retraining as product catalogs evolve, a crucial consideration for dynamic e-commerce environments.

Despite its strengths, our text-only approach has inherent limitations when classifying products with similar textual descriptions but different visual characteristics. To address this limitation, for future work, we plan to integrate image features and explore multimodal embedding approaches to further enhance classification accuracy, inspired by approaches like [10]. By combining textual and visual information, we anticipate significant improvements in distinguishing between visually distinct but textually similar products.

Additional promising directions include:

- 1) Exploring hierarchical classification approaches that explicitly model the relationships between top and bottom-level categories
- 2) Implementing attention mechanisms to focus on the most discriminative parts of product descriptions.
- 3) Incorporating customer behavior data such as search queries and browsing patterns to enhance classification performance
- 4) Developing semi-supervised learning techniques to leverage partially labeled product datasets, which are common in e-commerce settings
- 5) These extensions would build upon the solid foundation established in this work while addressing its limitations. The combination of interpretable models with multimodal data sources represents a particularly promising direction for practical e-commerce categorization systems that balance performance with explainability and deployment considerations

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